(a) A system of equations with exactly one solution

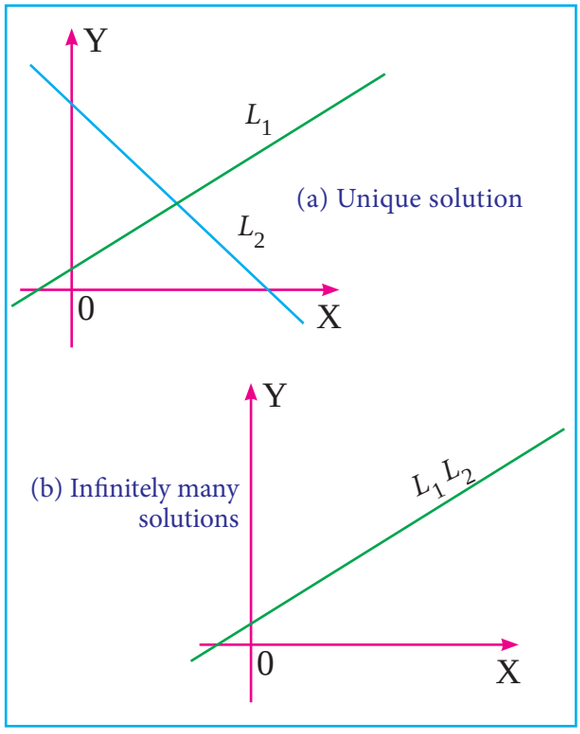
Consider the system $2x - y = 1$, $3x + 2y = 12$ which represents two lines intersecting at (2, 3) i.e (2, 3) lies on both lines. So the equations are consistent and have unique solution.

(b) A system of equations with infinitely many solutions

Consider the system $2x - y = 1$, $6x - 3y = 3$ which represents two coincident lines. We find that any point on the line is a solution. The equations are consistent and have infinite sets of solutions such as $(0, -1)$, $(1, 1)$ and so on. Such a system is said to be dependent.

(c) A system of equations that has no solution

Consider the system $2x - y = 1$, $6x - 3y = 12$ which represents two parallel straight lines. The equations are inconsistent and have no solution.



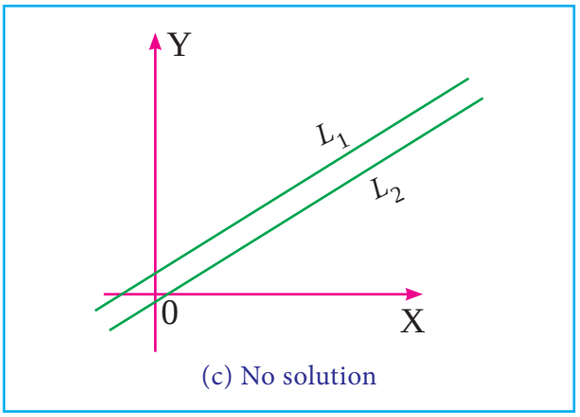


Fig. 1.1

System of non Homogeneous Equations in three variables

A linear system composed of three linear equations with three variables x, y and z has the general form

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \qquad (2)$$

A linear equation $ax + by + cz = d$ (a, b and c not all equal to zero) in three variables represents a plane in three dimensional space. Thus, each equation in system (2) represents a plane in three dimensional space, and the solution(s) of the system is precisely the point(s) of intersection of the three planes defined by the three linear equations that make up the system. This system has one and only one solution, infinitely many solutions, or no solution, depending on whether and how the planes intersect one another. Figure 1.2 illustrates each of these possibilities.

In Figure 1.2(a), the three planes intersect at a point corresponding to the situation in which system (2) has a unique solution.

Figure 1.2 (b) depicts a situation in which there are infinitely many solutions to the system. Here the three planes intersect along a line, and the solutions are represented by the infinitely many points lying on this line.

In Figure 1.2 (c), the three planes are parallel and distinct, so there is no point in common to all three planes; system (2) has no solution in this case.

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